SEAN KHOMPHENGCHAN

CURRENT ADDRESS
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SOCIALS
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OBJECTIVE

I am looking for Spring/Summer internship engineering positions that match my interest in GNC, Software, or Robotics.

EDUCATION

Purdue University, West Lafayette, IN — December 2023
Bachelor of Science in Aeronautical & Astronautical Engineering — GPA: 3.21 / 4.00
Specializations: Dynamics and Control
Relevant Coursework: Spacecraft Attitudes, Multi Agent Autonomy, Applied Controls in Astronautics, Satellite Navigation
Technical Skills: MATLAB, Python, C, C++, Simulink, Git, Stability Analysis, Kalman Filtering, Convex Optimization

TECHNICAL & RELEVANT EXPERIENCE

Undergraduate Research Assistant — October 2023 – Present
Purdue University, West Lafayette IN
- Working in the Autonomous & Intelligent Multi-agent Systems (AIMS) Lab for Space Robotics Logistics.
- Research current literature on possible control methods for spacecraft rendezvous and apply a Multi-Agent Extension.
- Using C/C++ to apply a hybrid of analytical and data driven control methods for optimization and safety guarantees.

Modeling and Simulation Intern — June 2022 – August 2023
Raytheon, Arlington VA
- Developed algorithms to predict and simulate platform performance and to conduct threat assessments.
- Briefed executives all the way to the president of the company on work accomplished and impact of.
- Directly contributed to a F-35 engine study that led to the winning of a $5.2-billion contract for an engine upgrade.
- Utilized Object Oriented Python and Java to build campaign level simulation software.
- Used Monte-Carlo Simulations to analyze effectiveness of different strategies and platforms.

Undergraduate Teaching Assistant — August 2021 – May 2023
Purdue University, West Lafayette IN
- Provided guidance to over 100 students on MATLAB, Python, and Excel to refine course concepts.
- Piloted a successful computer lab for students to access help for bridge programming and engineering skills.
- Assisted in the design of coursework and curriculum for First Year Engineering.

Undergraduate Research Assistant at Ma Quantum Lab — January 2021 – June 2021
Purdue University, West Lafayette IN
- Researched Microwave electronics for superconducting quantum circuits for use on quantum simulation testing.
- Accomplished noise reduction of 90% on a log curve when trying to analyze signal received from system.
- Collaborated with lab group to exchange ideas for finding better methods for end-to-end testing of system.

PROJECTS

Multi Agent Simulation for Competitive Esports — April 2023 – May 2023
- Built a program to simulate 5-unit vs 5-unit battles modeling a match within a Tactical Shooter Game.
- Created using Python and JavaScript utilizing the Mesa library.
- Able to verify accuracy by comparing against real life match data.

Convex Six-DOF Spacecraft Model — January 2022 – May 2023
- Incorporated Kalman Filters into control system design of active controlled model rocket.
- Create and model motion of spacecraft using sensor data with MATLAB and Simulink.
- Convexified trajectory and attitude of spacecraft using CVX library for reorientation.

Filtering Analysis for GNSS Signal Correction — August 2022 – December 2022
- Semester long research project in conjunction with Satellite Navigation Course.
- Utilizes a variety of different verified Filtering models for signal analysis of Satellite Navigation Systems in Python.
- From independently sourced GNSS data, was able to get an accuracy of 5 meters and low mean square root error.